Coordination Compounds: The Ultimate Revision Notes

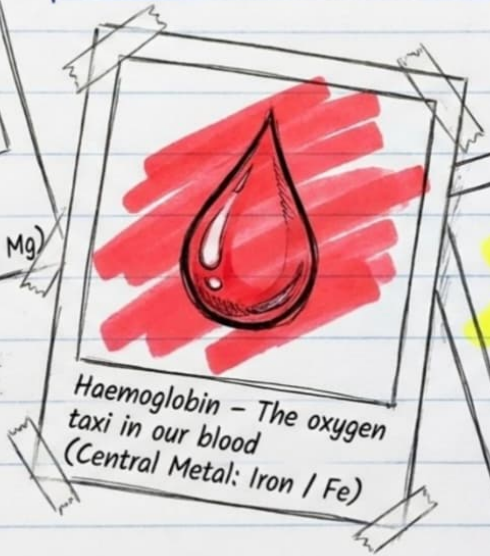
(A-Level / Uni Survival Guide!)

Exam prep! 🚀
Includes Werner's, Isomerism, VBT & CFT.

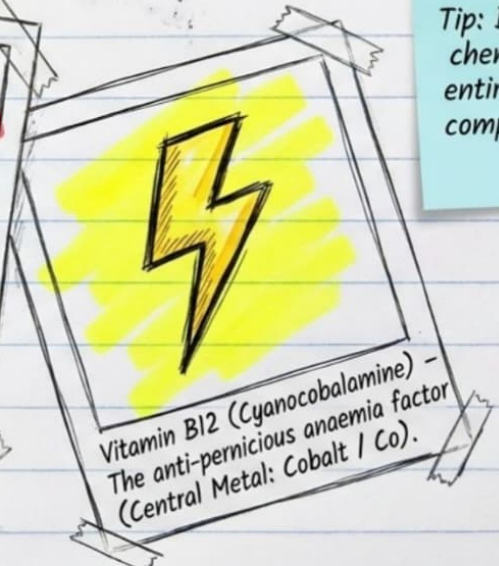
Nature's Secret Weapons

A central metal atom surrounded by a posse of molecules (ligands)! Transition metals are usually the stars of the show.

Chlorophyll – The green power in plants
(Central Metal: Magnesium / Mg)



Haemoglobin – The oxygen taxi in our blood
(Central Metal: Iron / Fe)



Vitamin B12 (Cyanocobalamine) – The anti-pernicious anaemia factor
(Central Metal: Cobalt / Co).

Tip: Bio-inorganic chemistry relies entirely on these complexes!

The Essential Vocab (Anatomy of a Complex)

Central Metal Ion:
The boss (Lewis Acid).
e.g., Co^{3+}

Ligands:
The electron donors.

Counter Ions:
The outer bodyguards.
Ionisable!

$$[Co(NH_3)_6]Cl_3$$

Coordination Sphere:
Everything inside is a single unbreakable unit!

Coordination Number (CN):
How many donor atoms are directly attached. Here, CN = 6.

Ligand Types

- Unidentate: 1 connection (e.g., Cl^-, H_2O)
- Didentate: 2 connections (e.g., 'en' - ethane-1,2-diamine)
- Polydentate/Chelate: Multiple connections (e.g., $EDTA^{4-}$ grabs the metal like a claw!).

⚠ Ambidentate ligands = 2 possible donor atoms but only use 1 at a time! (e.g., NO_2^- vs ONO^-).

Werner's Theory (The Blueprint)

Alfred Werner (1898)

	Primary Valence	Secondary Valence
1. Nature	Ionisable	Non-ionisable
2. Satisfied by	Negative ions	Neutral molecules or negative ions
3. Corresponds to	Oxidation State	Coordination Number
4. Geometry	Non-directional	Dictates spatial shape (polyhedron)

$$CoCl_3 \cdot 6NH_3$$

Primary!
(Precipitates as AgCl)

Secondary!
(Stays bonded to Co)

Summary:
Primary = Charge.
Secondary = Shape!

IUPAC Naming Rules (The Recipe)

- [x] **Step 1:** Cation first, Anion second (always!).
- [x] **Step 2:** Inside the [ ], name ligands alphabetically (irrespective of charge!).
- [x] **Step 3:** Add prefixes for quantity (di, tri, tetra). Use bis, tris for complex ligands!
- [x] **Step 4:** Name the metal. (If complex is an anion, add '-ate' e.g., Ferrate, Cobaltate).
- [x] **Step 5:** Add oxidation state in Roman numerals (III).

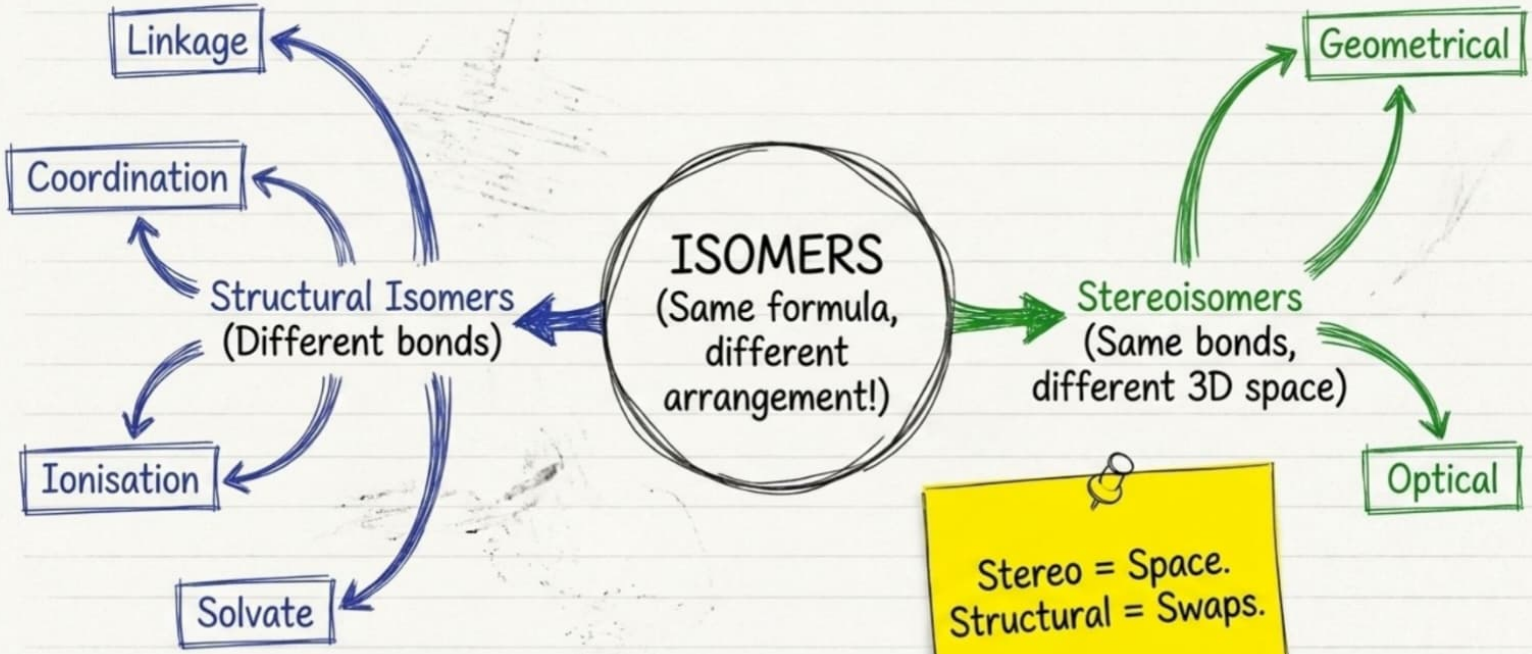
$[NiCl_2(PPh_3)_2]$

Breakdown: 2x Chloride -> dichlorido.
2x Triphenylphosphine -> bis(triphenylphosphine).
Metal -> nickel(II).

⇨ **Final Name:** dichloridobis(triphenylphosphine)nickel(II)

Spelling matters!
Aqua = H_2O.
Ammine = NH_3
(Two M's!!)

Isomerism (The Family Tree)



Stereo = Space.
Structural = Swaps.

Valence Bond Theory (VBT) & Shapes

Hybridisation	Shape
sp3	Tetrahedral (CN=4)
dsp2	Square Planar (CN=4)
sp3d2	Octahedral (CN=6)
d2sp3	Octahedral (CN=6)

Inner vs Outer

Inner Orbital Complex:
Uses the (n-1)d orbitals.
Example: d2sp3.
(Usually low spin).

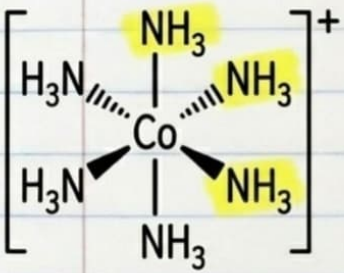
Outer Orbital Complex:
Uses the nd orbitals.
Example: sp3d2.
(Usually high spin).

Magnetic property tip:
Unpaired electrons = Paramagnetic.
All paired = Diamagnetic.

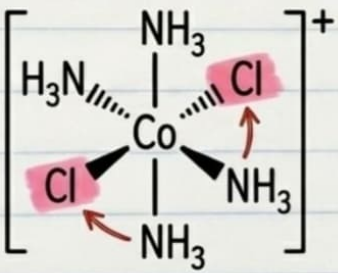
Stereoisomers (Drawing 3D in 2D)

Geometrical Isomerism

Depends on angles! Cis = adjacent (90°). Trans = opposite (180'). Fac/Mer for Ma_3b_3.



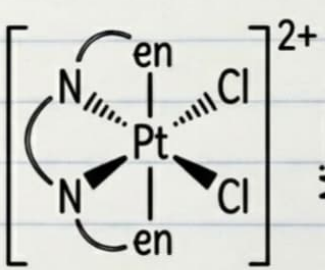
Cis-isomer



Trans-isomer

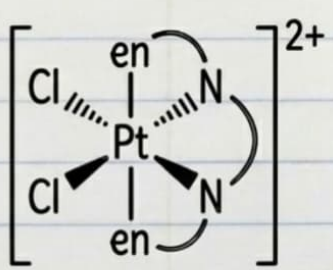
Optical Isomerism

Non-superimposable mirror images (Chiral). Dextro (d) and Laevo (l).



Dextro (d) isomer

Mirror

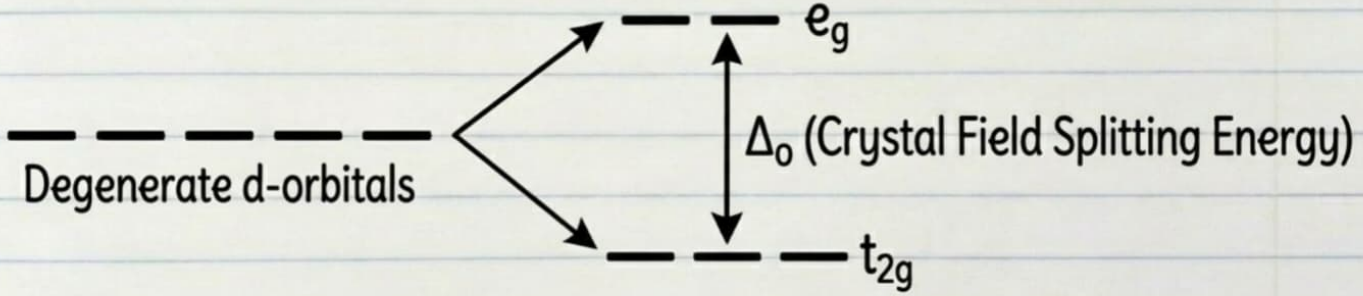


Laevo (ℓ) isomer

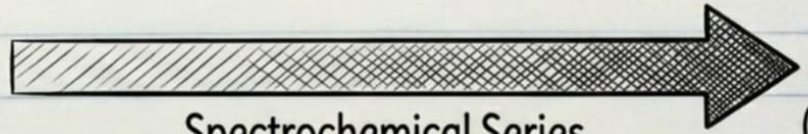
✳ *Trans* isomers are usually too symmetric to have optical isomers. Always check *Cis*!

Crystal Field Theory (CFT) - The Splitting

Ligands approach the metal → electrons repel each other → the 5 degenerate d-orbitals split into two distinct energy levels!



Weak field:
I^-, Br^-, H_2O
(Produces small Δ_o gap)

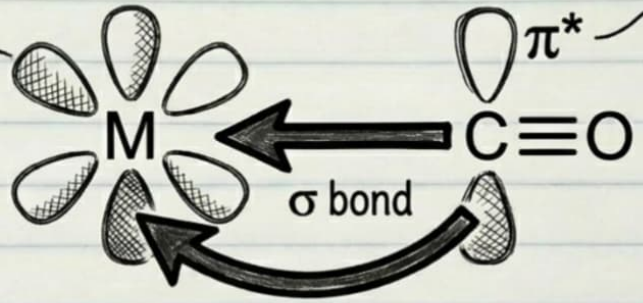


Strong field:
CN^-, CO
(Produces huge Δ_o gap)

Metal Carbonyls & Synergic Bonding

It's the ultimate chemical teamwork!
A two-way electron street that makes the bond super strong.

The Carbonyl ligand (CO) donates a lone pair of electrons into the Metal's empty orbital.



The Metal donates electron density from a filled d-orbital back into the CO's empty anti-bonding π* orbital.

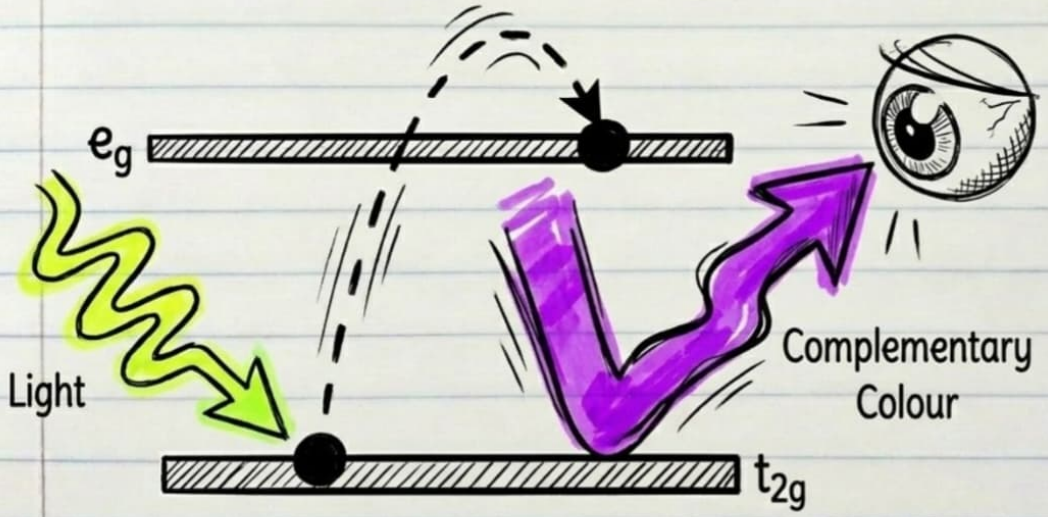
Result: This mutual reinforcement is called the **Synergic Effect.**

✳ Carbonyls (CO) are homoleptic — metal oxidation state is usually ZERO! e.g., $[Ni(CO)_4]$

Why are they so colourful? (d-d Transitions)

1. The complex absorbs a specific wavelength of visible light.
2. A lower t_{2g} electron gets excited and physically jumps the Δ_o gap up to e_g.
3. The colour we actually see is the complementary colour that bounced off!

No unpaired d-electrons? No d-d transition = Colourless! (e.g., Zn^{2+})



$[Ti(H_2O)_6]^{3+}$ absorbs yellow/green light, so the solution appears Violet! 💜

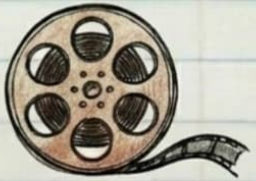
Why It Matters (The Real World)

Medicine

- Cis-platin: Effectively inhibits the growth of tumours (anti-cancer).
- Chelate Therapy: EDTA is used to treat lead poisoning by grabbing the toxic metal. D-penicillamine removes excess copper/iron.

Industry & Metallurgy

- Wilkinson's Catalyst: A Rhodium complex used for hydrogenation of alkenes.
- Electroplating: Complexes like $[Ag(CN)_2]^-$ allow for much smoother plating of silver and gold.



Photography

- Black & White Film: 'Hypo' solution dissolves undecomposed AgBr to fix the film, forming the complex ion $[Ag(S_2O_3)_2]^{3-}$.

THE MASTER CHEAT SHEET

THE COORDINATION COMPLEX

dictated by →

Werner's Theory (Primary = charge, Secondary = shape/CN).

named using →

IUPAC Rules (Ligands alphabetical, prefixes, oxidation oxidation states).

structure explained by →

VBT & Isomerism (Hybridisation, 3D spatial geometry, swaps).

colour/magnetism explained by →

Crystal Field Theory (Δ_o gap, d-d transitions, strong/weak fields).

You've got this! Good luck on the exam! 🍀